DETERMINANTS · GEOMETRY

What the determinant measures, by dimension

The magnitude is a volume, the sign is an orientation, and zero is a collapse. Only the word for "volume" changes as the dimension rises — which is the argument for calling the general case a volume at all.

5
DIMENSIONS

WHERE THE PICTURE IS AVAILABLE 3

1 **One dimension**

`length scaling; sign = direction`

$\det[a] = a$

A segment on a line. The determinant is the number itself, and its sign says whether the direction is preserved or reversed — the degenerate case that makes the general statement believable rather than surprising.

2 **Two dimensions**

`sign: counterclockwise +, clockwise −`

$|\det A|$ = area of the parallelogram

The columns of A span a parallelogram, and the determinant is its signed area. Zero means the columns are parallel and the parallelogram has collapsed to a line — which is the same statement as the columns being linearly dependent.

3 **Three dimensions**

`sign: right-handed +, left-handed −`

$|\det A|$ = volume of the parallelepiped

Three columns span a parallelepiped. The sign distinguishes a right-handed frame from a left-handed one — the same distinction the cross product encodes, and the reason $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ is a determinant.

WHERE IT IS NOT 2

4 **Four dimensions and above**

`orientation preserved + or reversed −`

$|\det A|$ = volume of the image of the unit cube

No picture, and none needed — the algebra is unchanged. This is where the determinant stops being a description of something already visible and becomes the definition of n-dimensional content, which is what the change of variables formula then relies on.

5 **Determinant zero**

`image lies in a proper subspace`

$\det A = 0$

The collapse case in every dimension: the image has zero content because it fits inside something lower-dimensional. Note the determinant reports that a collapse happened but not by how much — a matrix flattening $\mathbb{R}^3$ to a plane and one flattening it to a point both give zero. That is what rank is for.

Read down the column and the pattern is one definition stated four ways: $|\det|$ is the factor by which the unit cube's content is scaled, the sign says whether the image is reflected, and $\det = 0$ says the image has no content because it fits inside a smaller subspace. Length, area and volume are not analogies here — they are the $n = 1, 2, 3$ cases of the same quantity.